Prediction of Water Quality Index in Great Lakes

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***Abstract*—In this paper, a variety of methods are used to predict the Water Quality Index of areas in the Great Lakes region. Using a Multiple Linear Regression (MLR) model, a Support Vector Regression (SVR) model, and a Neural Network to predict the water quality index showed that the neural network generally performed better at the task than the MLR and SVR models, with improvements in the R^2 and RMSE values over both other models, as well as an improvement in MAE over the MLR model. Also identified were the most important predictors for the water quality index: oxygen, water temperature, chloride, fluoride, and chlorophyll a (uncorrected for pheophytin). Additionally, a time series model was used to predict the next water quality index in sequence. To this end, a Long Short-term Memory model (LSTM) was used, whose overall performance was better than the MLR model's, but worse than the other two attempted models.**

***Index Terms*—water quality index, machine learning (ML), neural networks (NNs), Great Lakes**

I. Introduction

The Great Lakes are an important part of the water supply and environment of the USA and Canada. As such, being able to accurately predict their water quality is important in ensuring their health. To this end, we will design a variety of different machine learning models capable of predicting the water quality index of the Great Lakes based on water quality data from the Great Lakes.

With these models, sudden changes in the Great Lakes' water quality can be caught quickly, allowing action to be taken to correct the change just as quickly. With this rapid response time, the water quality, which the ecosystem in the Great Lakes relies on, can be more efficiently preserved.

II. Related Work

Miller et al. used both statistical and ML approaches to predict the water quality, using data collected over a span of 20 years. They implemented Multiple Linear Regression (MLR), Artificial Neural Network (ANN), and a Random Forest (RF) algorithm to predict the water quality. Their study showed that the RF algorithm had the best performance, and also showed the importance of Biological Oxygen Demand (BOD) and Turbidity to predict water quality [1].

Hussein et al. used the classification learner tool in MATLAB to choose the best decision tree model among support vector machines (SVM), k-nearest neighbors (K-NN), ensemble tree (ET), and discrimination analysis (DA). Their results indicate that linear SVM performed the best, and can predict the water quality index with good accuracy [2].

The study by Trebitz et al.(2007) analyzed the water quality of various 58 coastal areas in the Great Lakes basin where they analyzed the agriculture disturbance, nutrient concentrations, and suspended particulates and found that Lake Erie wetlands showed greatest nutrients but lowest clarity. They used the Great Lakes Ecological Indicators consortium(GLEI) method. A PCA based index summarized patterns perfectly, but other type classification failed to explain variation [3].

While Miller et al. did create a means to predict water quality, they did so on a reservoir in Poland [1]. We build on this by doing the same for the Great Lakes, which are much larger bodies of water. Similarly, Hussen et al. performed water quality prediction on groundwater [2], so we will perform a similar process using the Great Lakes instead. Trebitz et. al. did use Great Lakes data, but calculated the water quality using the samples they collected instead of using an ML approach [3]. We will be using an ML approach to predicting water quality, but without the data Trebitz had access to, we use the "Great Lakes Water Quality Monitoring and Surveillancce Data" provided by Datastream instead [5].

III. Data Description

The dataset we have used in this study is collected by Environment and Climate Change Canada (ECCC) [5], and was last updated on 2025-09-22. The dataset contains numerous measurements of various factors affecting the water quality in the Great Lakes region. The data currently collected spans 24 years, starting from 2000 up to 2024 in the latest version.

The original dataset consists of roughly 237k observations, but it is formatted such that each measurement has multiple features. The original dataset contained all the predictors in the *CharacteristicName* column, and the values in the *ResultValue* column. We pivoted the dataset such that the columns are the predictor variables.

After this transformation, there were 6550 rows with 62 predictors (continuous variables) and 3 indices (categorical variables). After filtering columns with null values, the dataset had 18 predictors, which were mostly right-skewed. The final dataset had 6550 observations, and 22 columns among which 18 were predictors, 3 were index variables, and 1 was the target variable.

IV. METHODOLOGY

A. Data Pre-Processing

We used a standard data pre-processing pipeline to ensure that the dataset is fit for training our models. The summary is as follows:

- Dropped fully null columns, and columns with more than half missing
- Applied KNN imputation
- Calculate WQI indices and created the target column
- Skewness correction with Box-Cox
- Outlier detection and winsorization

The dataset is quite sparse, and thus all the null columns, along with columns with greater than half missing, were dropped. After this, the predictors were scaled using the *StandardScaler()* and the missing values were imputed using KNN imputation ($n_neighbors = 10$).

After this, the WQI was calculated for each row using the weighted formula,

$$WQI = \frac{\sum_{i=1}^{n} w_i \cdot q_i}{\sum_{i=1}^{n} w_i} \tag{1}$$

where w_i is the weight of the i-th parameter, q_i is the quality rating of the i-th parameter, and n is the number of parameters. All the predictors are assigned the same weight and contribute equally to the model. The quality rating is the relative concentration of the measured value with respect to its standard limit. The standard values were compiled from commonly cited water quality guidelines, including United States Environmental Protection Agency (EPA) water quality criteria, United States Geological Survey (USGS) reference values, and other widely used environmental monitoring thresholds as reported in literature [6], [7], [8], [9], [10], [11], [12], [13], [14], [15], [16], [17].

This equation produces a single number, which possesses thresholds for water quality. According to the University of South Florida, a WQI of 0 to 45 means "Good" water quality, 45 to 60 means "Fair" water quality, and greater than 60 means "Poor" water quality [17].

The dataset had a right-skewed distribution, which was corrected using Box-Cox transformation. Using Z-score, roughly 7% of the dataset was classified as outlier, with a Z-score greater than 3 or less than -3. The effect of outliers was minimized using winsorization. No correlation filtering was applied, even though some predictor-pairs had high correlation.

After all the preprocessing steps, we had the following dataset:

- 6550 observations
- 17 predictor columns, 3 index columns
- 1 target variable column (WQI)

Since the dataset only contains about 6500 observations, we decided to use a smaller testing and validation split. We created a 90/5/5 split for training, testing and validation sets respectively.

The *ActivityStartDate* column has measurements that are taken at random intervals. For time-series analysis, we smoothed out the WQI using a rolling average (window size 3) such that the outliers in WQI are minimized.

B. Models implemented

1) Linear Regression: We started with the simple Multiple Linear Regression (MLR) model using the *sklearn* library.

We used 5-fold cross-validation to fit the model and extracted the cross-validation results.

2) Support Vector Regression: Support Vector Regression (SVR) model was implemented using the Radial-Basis Function (RBF) kernel.

We used 5-fold cross-validation to fit the model across a grid of (C, $epsilon$, $gamma$) and extracted the best model.

3) Neural Network: A Neural Network (NN) model was constructed with the following parameters:

- 1 input layer
- 3 hidden layers, ReLU activation
- 1 output layer, linear activation
- L2 regularization, Xavier Normal initialization
- Adam optimizer, learning rate = $1 \cdot 10^{-3}$
- Mean Squared Error loss function
- 100 epochs

The model was created using the same process used to create multi-layer models in the class assignments A1, A2, and A3 [18], [19], [20].

C. Time-Series Forecasting

We implemented a univariate model, considering only the WQI values for simplicity. A Long Short-term Memory (LSTM) model was constructed using a new 90/10 split. The features in the dataset were scaled using *MinMaxScaler()*, and sliding window sequences were used to feed previous datapoints to the model. The LSTM was trained with the following parameters:

- 1 LSTM layer with 16 units
- Adam optimizer, learning rate = $5 \cdot 10^{-4}$
- 0.1 validation split
- 200 epochs, with early stopping
- 4 batch size

D. Model evaluation

In order to evaluate the models, we used the standard metrics - Coefficient of determination (R^2), Root-Mean Squared Error ($RMSE$) and Mean Absolute Error (MAE).

We calculated these metrics using the results from the test set, and then plotted the predicted WQI against true WQI values. For the neural network and LSTM, we also plotted the Loss and Accuracy curves to evaluate the training process.

V. EXPERIMENTAL RESULTS

The cross-validation results reported in Table I suggests that the SVR model had the better performance when compared to the MLR model. However, the test-set results reported in Table II show that the neural network model had the best

performance among all the models tested. If we consider MAE as the testing metric, then the SVR model comes on top.

TABLE I
CROSS-VALIDATION RESULTS OF MLR AND SVR MODELS

Model	R^2	RMSE	MAE
MLR	0.7864	3.0003	1.7542
SVR	0.9646	1.2383	0.2645

TABLE II
TESTING RESULTS OF MLR, SVR, NN, AND LSTM MODELS

Model	R^2	RMSE	MAE
MLR	0.8165	2.4582	1.5674
SVR	0.9595	1.1551	0.2738
NN	0.9779	0.8525	0.4560
LSTM	0.8749	1.2599	0.8288

We also used the *permutation_importance* function in the *sklearn* library to determine the most important predictors in each model. The results are listed in the Tables III through V. The actual importance values for the predictors are not reported here.

The most important predictors across all the models can be extracted as follows (not ranked by importance):

- Dissolved Oxygen (DO)
- Temperature, water
- Chloride
- Fluoride
- Chlorophyll a, uncorrected for pheophytin

TABLE III
TOP 3 IMPORTANT PREDICTORS FOR MLR MODEL

Rank	MLR
1	Dissolved oxygen (DO)
2	Temperature, water
3	Chlorophyll a, uncorrected for pheophytin

TABLE IV
TOP 3 IMPORTANT PREDICTORS FOR SVR MODEL

Rank	SVR
1	Chloride
2	Fluoride
3	Dissolved oxygen (DO)

The comparison between the four models reveals a clear difference in their prediction capability. The MLR model performed an R^2 of 0.8165, RMSE of 2.4582, and MAE of 1.5674 on the test set, which indicate moderate predictive skill. The SVR model performed much better than the MLR with an R^2 of 0.9595 and RMSE of 1.1551 with the lowest MAE of 0.2738. The Neural Network achieved the best accuracy overall with R^2 of 0.9779 and the lowest RMSE of 0.8525, but its MAE (0.4560) was higher than that of SVR. Finally, the LSTM model performed better than the MLR model, but worse than the SVR and NN models, with an R^2 of 0.8749, an RMSE of 1.2599, and a MAE of 0.8288.

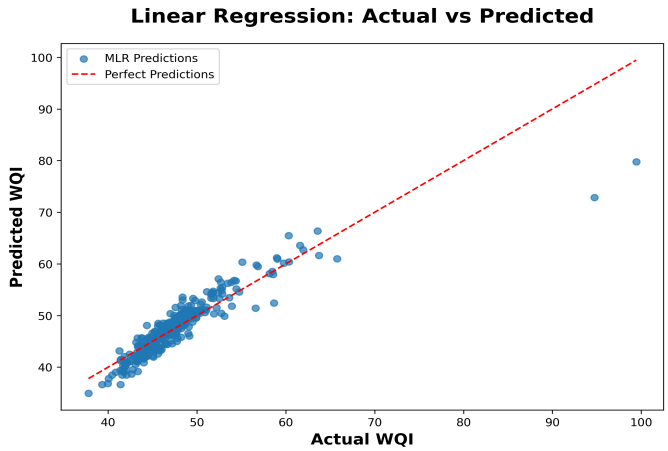

Fig. 1. Linear regression: Actual vs Predicted WQI.

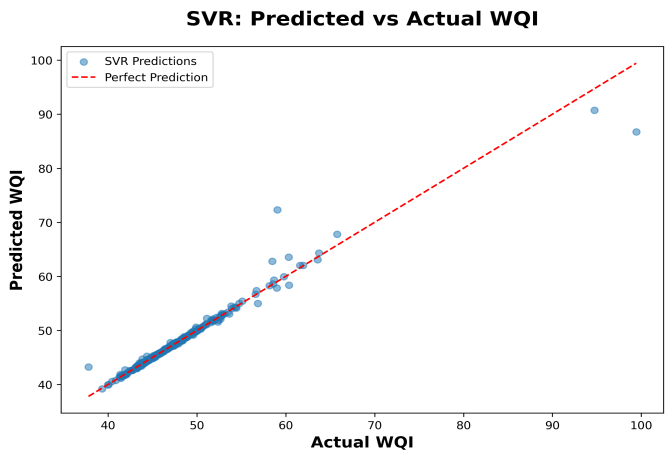

Fig. 2. Support vector regression: Actual vs Predicted WQI.

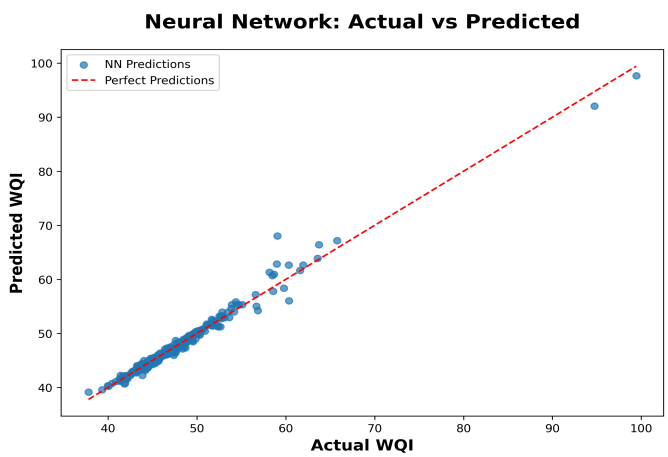

Fig. 3. Neural network: Actual vs Predicted WQI.

TABLE V
TOP 3 IMPORTANT PREDICTORS FOR NN MODEL

Rank	NN
1	Chloride
2	Dissolved oxygen (DO)
3	Temperature, water

The quantitative results in Figure 1 are consistent with the qualitative pattern observed in the prediction plot. The MLR predictions highlight significant depression around the 1:1 reference line, highlighting systematic deviations from actual WQI values. This is further supported by the residual distribution plot (Fig. 2), where the majority of errors lie within the range of ±5 units. This illustrates that, while MLR captures the general trend, it somehow lacks some flexibility to model nonlinear relationships in the data.

The SVR model (Fig. 3) demonstrates a tighter clustering of predictions along the ideal line, indicating its strong performance in minimizing average error. However, some random points deviate significantly from the trend. Although the numbers are small, this explains that despite its low MAE, its RMSE and R^2 are slightly lower than the Neural Network. This reflects that SVR is efficient in capturing the structure of the data but is more sensitive to local prediction errors.

The Neural Network results (Fig. 4) demonstrate the closest alignment between the predicted and actual WQI values, with the closest dispersion across the full range. Its superior R^2 and RMSE performance highlights the consistency of this model. The training and validation curves (Fig. 5 and Fig. 6) demonstrate an accurate and consistent decrease in error over epochs, indicating stable convergence. The fluctuations in the validation curve are consistent with the small validation datasets, rather than instability in the model.

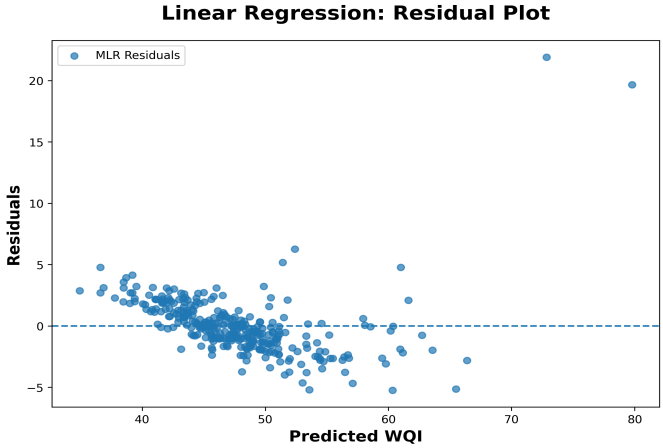

Fig. 4. Linear regression residuals.

The training and validation error curves were plotted for the neural network using a log-scale for the Y-axis, which showed that the NN trained well and performed optimally on the validation set. The spikes on the validation curves are most likely caused by the small validation set.

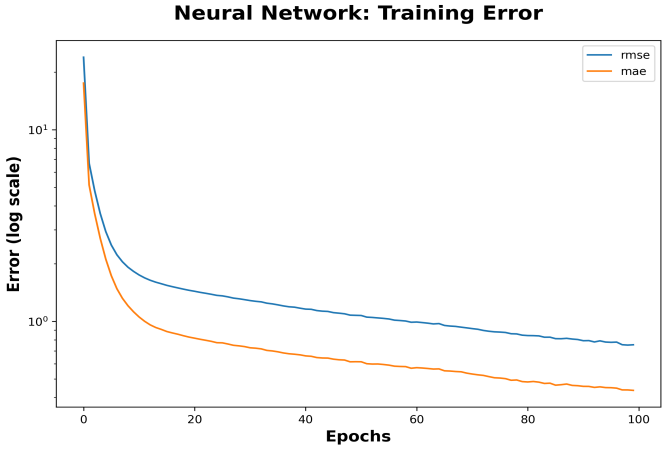

Fig. 5. Neural network training errors.

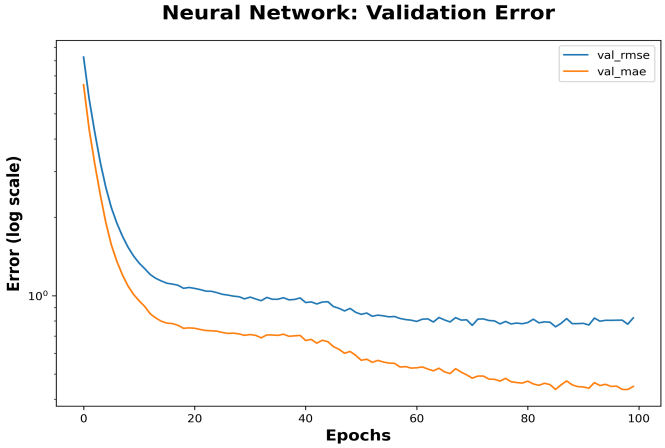

Fig. 6. Neural network validation errors.

VI. DISCUSSION AND CONCLUSION

This study implemented the four models - MLR, SVR, NN and LSTM. The performance of these models illustrate the capability of machine learning approaches to analyze the relationship between water quality parameters and the Water Quality Index (WQI) in the Great Lakes basin datasets.

The Neural Network's high performance, compared to the other three models, shows the benefits ML can provide to the task of predicting Water Quality Indices. With the neural network model, Water Quality Index predictions can be generated quickly with high accuracy, which informs professionals quickly about any predicted change ahead of time, allowing them to correct the change and preserve the health of the Great Lakes efficiently.

It is clear that the preprocessing pipeline significantly influenced the performance of the model. The result across MLR, SVR, and NN indicates that these parameters are the key variables in determining WQI within the datasets and support the modeling approach. The LSTM model also showcased the ability to forecast WQI with good accuracy. The overall results highlight that the nonlinear models provide substantial

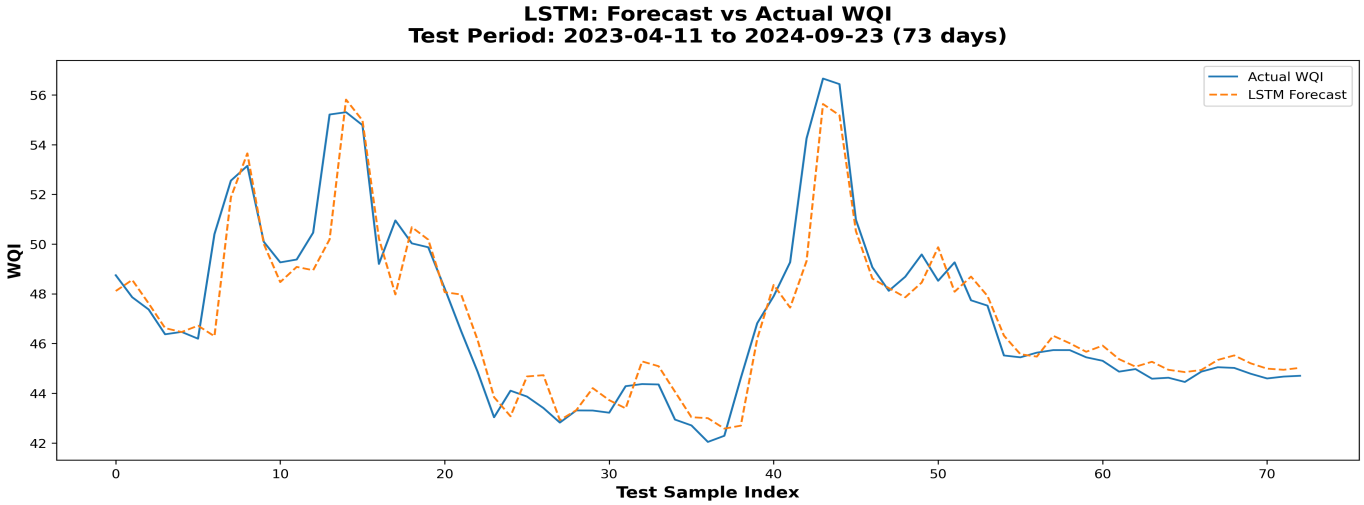

Fig. 7. LSTM: Actual vs Predicted WQI.

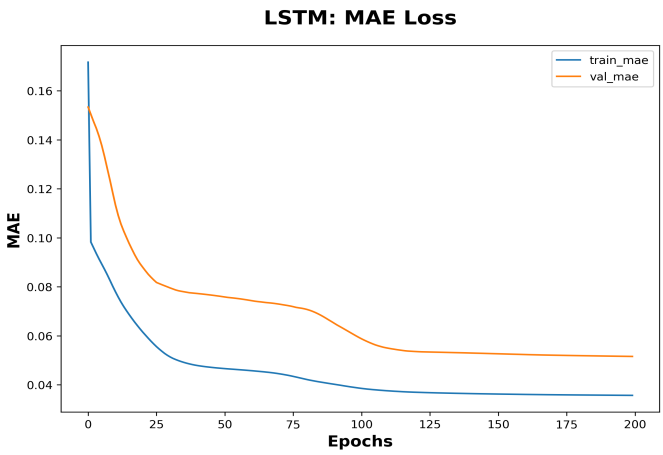

Fig. 8. LSTM Mean Absolute Errors: Train and Validation.

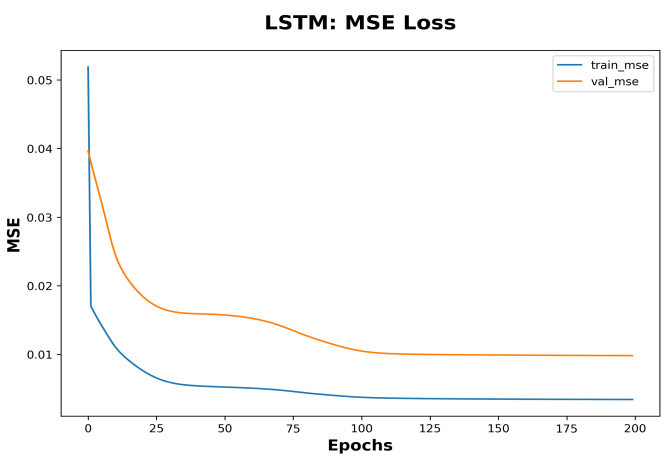

Fig. 9. LSTM Mean Squared Errors: Train and Validation.

improvements over linear methods for error reduction, and approximate the complex relationships between the predictors better.

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